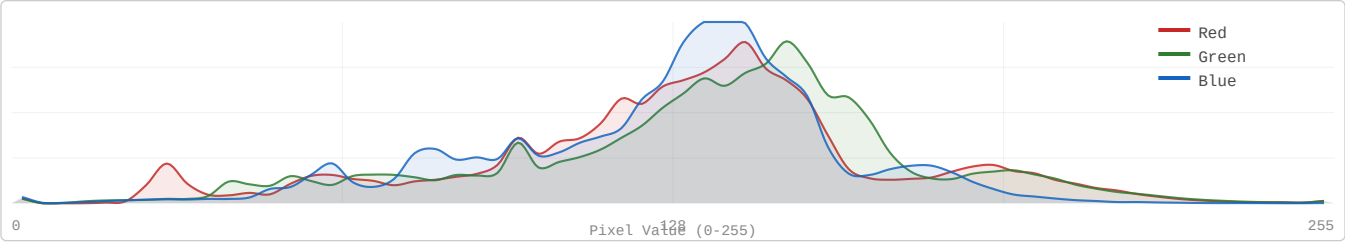
Decorrelation Gap: KODIM09

512x768 | 24-bit RGB | PNG (lossless)

Reeds and Marsh at Sunrise | Weakly one-dimensional (PC1 = 90.85%)

1. RGB Channel Distribution



2. Inter-Channel Correlation: RGB vs YCbCr

RGB (Original)

	R	G	B
R	1.0000	0.9486	0.7330
G	0.9486	1.0000	0.8560
B	0.7330	0.8560	1.0000

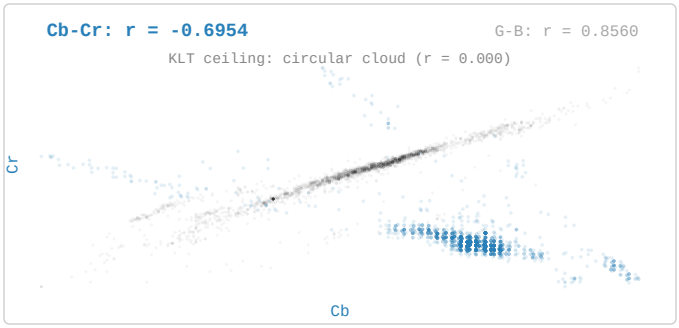
Cb-Cr: -0.695

(KLT ceiling = 0.000)

YCbCr (BT.601)

	Y	Cb	Cr
Y	1.0000	-0.4896	0.1209
Cb	-0.4896	1.0000	-0.6954
Cr	0.1209	-0.6954	1.0000

3. Cb vs Cr Scatter with G-B background overlay



4. Pairwise Decorrelation Gap

Channel Pair	RGB	YCbCr (BT.601)	Residual |r|
R-G → Y-Cb	0.948560	-0.489630	0.4896
R-B → Y-Cr	0.732954	0.120911	0.1209
G-B → Cb-Cr	0.856028	-0.695387	0.6954
Avg |r|	0.845847	0.435309	0.4353

Decorrelation Gap: KODIM09

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Reeds and Marsh at Sunrise | Weakly one-dimensional (PC1 = 90.85%)

5. Decorrelation Efficiency & Blue Independence

Decorrelation Efficiency

48.5% Moderate

0% BT.601 leaves all — 100% KLT ceiling

Blue Channel Independence (Baetzel 2026a)

22.0%

0% Fully predictable — 100% Fully independent

6. 4:2:0 Chroma Subsampling — Information Cost

Channel	PSNR (dB)
R	46.32 dB
G	49.79 dB
B	42.99 dB
Overall	45.52 dB

Method: BT.601 RGB→YCbCr | 2x box downsample Cb,Cr | nearest upsample | BT.601 inverse

7. PCA Baseline Reference (Baetzel 2026a)

Eigendecomposition

Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.6232	-0.6258	-0.4690	3928.74
PC2	0.5660	0.0530	-0.8227	353.13
PC3	-0.5398	0.7781	-0.3211	42.38

8. PCA Derived Metrics

Condition Number

92.70

PC1/PC2: 11.13 | PC2/PC3: 8.33

Blue Channel Independence

22.0%

Variance not captured by PC1

PC1 Dominant Channel

G

Loading: G (0.63)

9. Redundancy Profile

Classification: Weakly one-dimensional | CN: 92.70 | PC1: 90.8%

RGB Correlation: Avg |r| = 0.845847 | R-G: 0.9486 | R-B: 0.7330 | G-B: 0.8560

YCbCr Residual: Avg |r| = 0.435309 | Y-Cb: -0.4896 | Y-Cr: 0.1209 | Cb-Cr: -0.6954

Efficiency: 48.5% of KLT ceiling | Gap: 0.4353 avg |r| remaining

Blue Channel: 22.0% independent | 4:2:0 PSNR: 45.52 dB

References

[1] Baetzel, J. (2026a). "Statistical Characterization of Inter-Channel Redundancy Structure"
[2] Wallace, G.K. "The JPEG still picture compression standard." IEEE Trans. CE 38(1), 1992.
[3] ITU-R Recommendation BT.601: Studio Encoding Parameters, 1982.
[4] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.